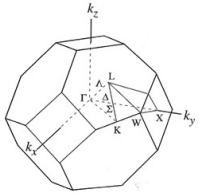
Metal-Oxide-Semiconductor



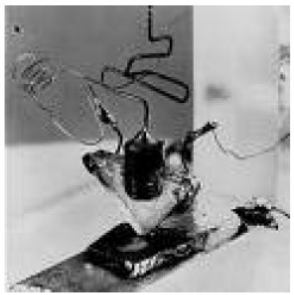
Scaling of MOSFET

Vacuum Tubes



1906-1950s

Bipolar



1947-1980s

MOSFET



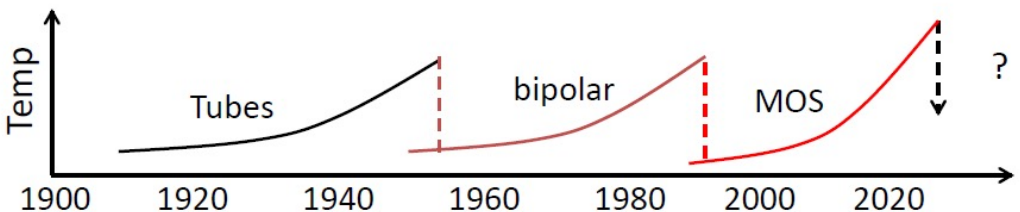
1960-until now

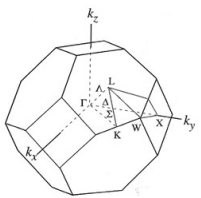
Now ??

Spintronics

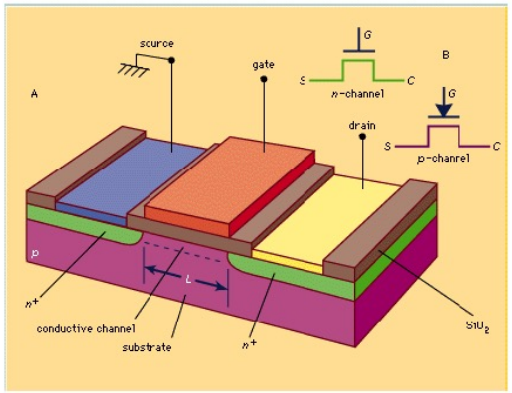
Bio Sensors

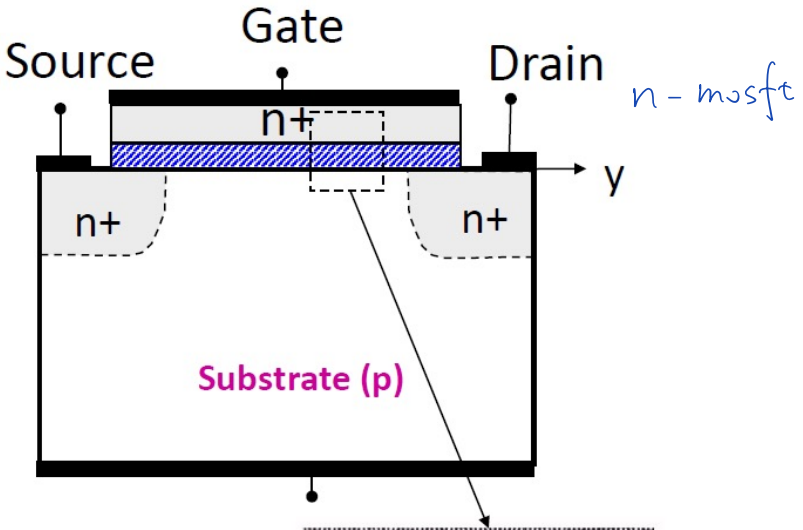
Displays ….

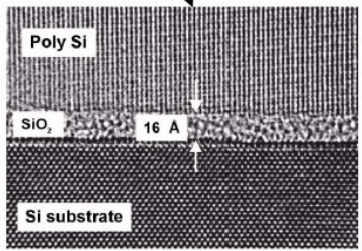




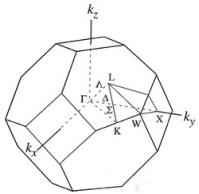
Basic Configuration of a MOSFET



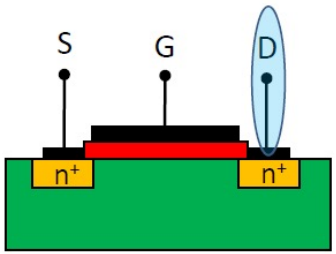


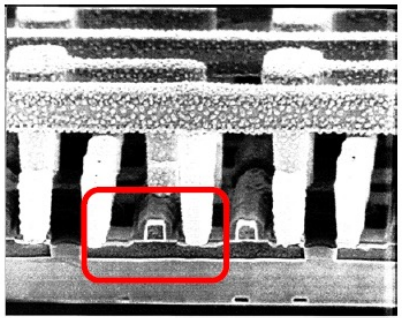


Almost like a lateral bipolar transistor!

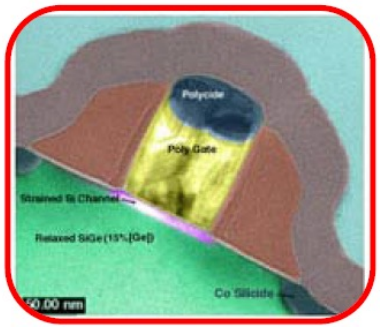


Background

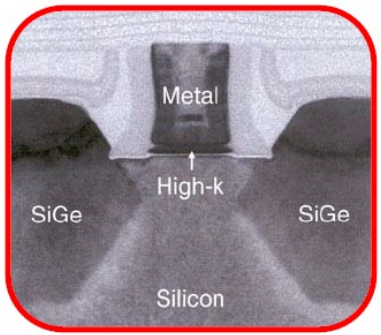




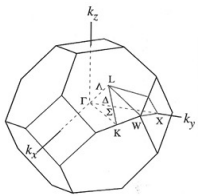
Strained MOSFET



High-k/metal gate MOSFET



Sources:
IBM J. Res. Dev.
Google Images
Intel website



Basic MOS Capacitor

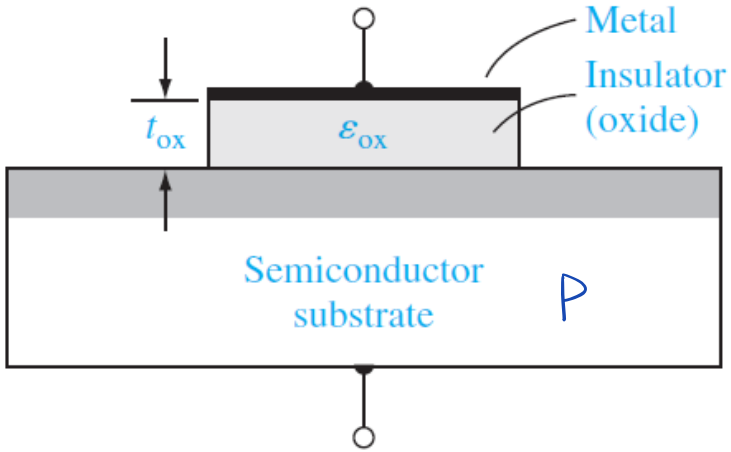
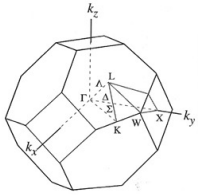
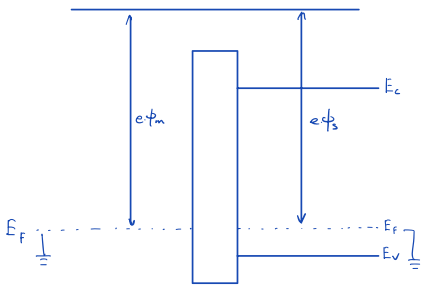


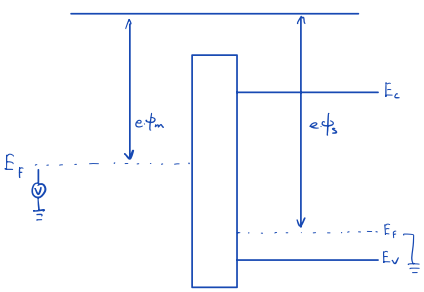
Figure 10.1 | The basic MOS capacitor structure.

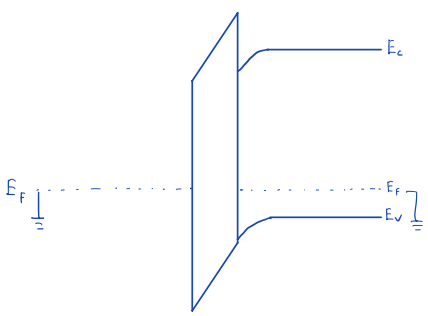


Assume $\phi_m = \phi_s$



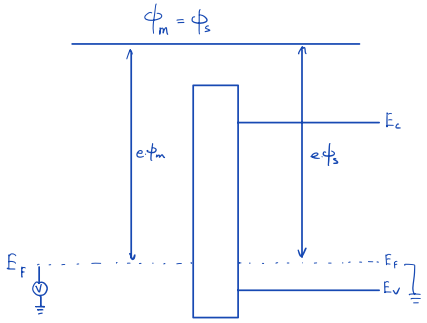
Assume $\phi_m < \phi_s$



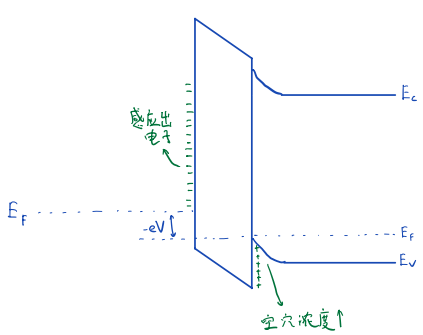


$V_{flat} = \phi_m - \phi_s = \phi_{ms}$

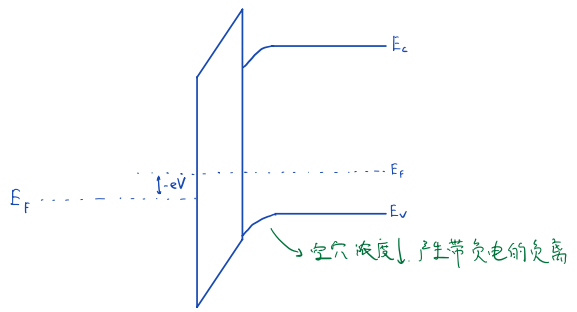
$\phi_m = \phi_s$

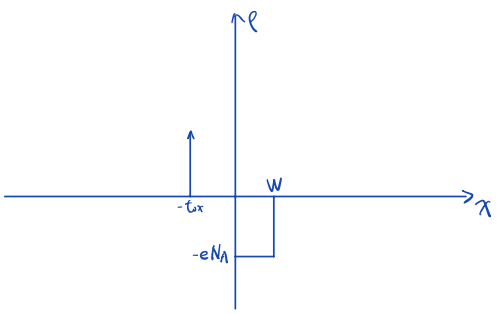


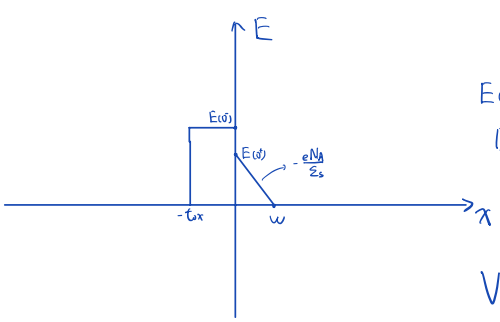
$V < 0$



$V > 0$



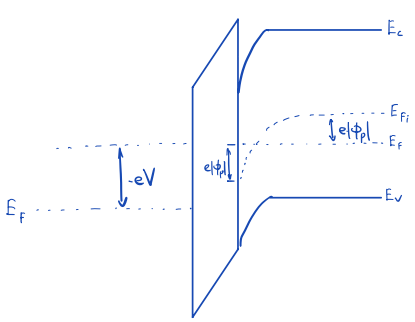




$$E(0) = \frac{eN_A}{\varepsilon_s} W$$

$$D(0^-) = D(0^+) \Rightarrow E(0^-) = \frac{eN_A}{\varepsilon_{ox}} \cdot W$$

$$V_G = \int_{-\infty}^{+\infty} E\,dx = V_{ox} + V_s = \frac{eN_A}{\varepsilon_{ox}} \cdot W \cdot t_{ox} + \frac{1}{2} \frac{eN_A}{\varepsilon_s} \cdot W^2$$



when $V_s = 2|\phi_p| \Rightarrow$ inversion

在表面处的电子浓度与半导体体内的空穴浓度相等

$V_G = V_{th}$. $\frac{eN_A}{\varepsilon_s} \cdot \frac{W^2}{2} = 2|\phi_p|$

$$W = 2\sqrt{\frac{|\phi_p| \varepsilon_s}{eN_A}} \quad . \quad V_{ox} = W \cdot \frac{eN_A \cdot t_{ox}}{\varepsilon_{ox}} = \frac{2\sqrt{eN_A |\phi_p| \varepsilon_s}}{\varepsilon_{ox}} \cdot t_{ox}$$

$$\boxed{V_{th} = V_G = \frac{2\sqrt{eN_A |\phi_p| \varepsilon_s}}{\varepsilon_{ox}} \cdot t_{ox} + 2|\phi_p| = \frac{Q_{ss}}{C_{ox}} + 2|\phi_p|}$$

$Q_{ss} = eN_A \cdot W$

$V_{Flat} = 0$

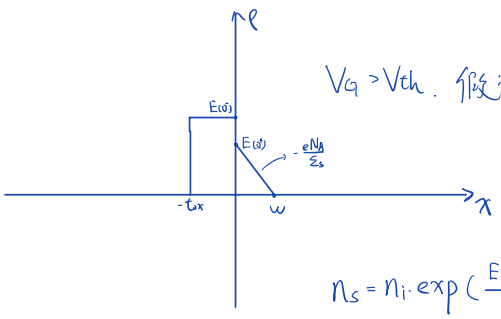
$$\boxed{V_{Flat} \neq 0 \quad . \quad V_{th} = \frac{Q_{ss}}{C_{ox}} + 2|\phi_p| + V_{Flat}}$$

$V_G > V_{th}$. $V_S \cong 2|\phi_p|$. W不发生改变

$$V_{th} = \frac{Q_{ss}}{C_{ox}} + 2|\phi_p|$$

$$V_G = \frac{Q_{ss} + Q_{inv}}{C_{ox}} + 2|\phi_p|.$$

$$Q_{inv} = (V_G - V_{th}) C_{ox}.$$



$V_G > V_{th}$. 假设W线性↗. V_S 平方增涨

$$E_{Fi}^s - E_{Fi}^b = -eV_s$$

$$n_s = n_i \cdot \exp\left(\frac{E_F - E_{Fi}^s}{kT}\right) = n_i \exp\left(\frac{E_F - E_{Fi}^b + eV_s}{kT}\right)$$

n_s：表面处

$$E_{Fi}^s - E_{Fi}^b = -eV_s$$

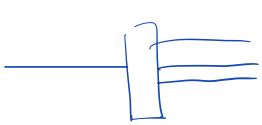
$$\Rightarrow n_s = n_b \exp\left(\frac{eV_s}{kT}\right)$$

n_b：体材料.

$V_{ox} = \frac{Q_{ss}}{C_{ox}}$. 指数↗. $V_G = V_{ox} + V_s$. 指数↗. 与 V_G 线性矛盾

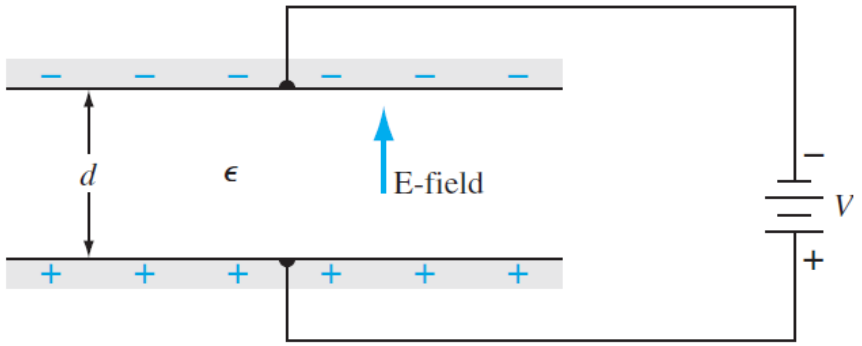
$\Rightarrow$ W 机乎不变

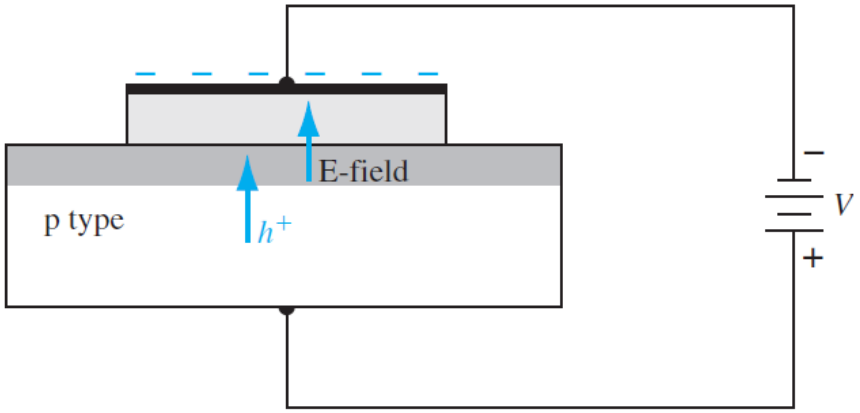


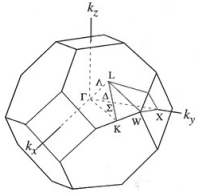


MOS with Negative Gate Bias

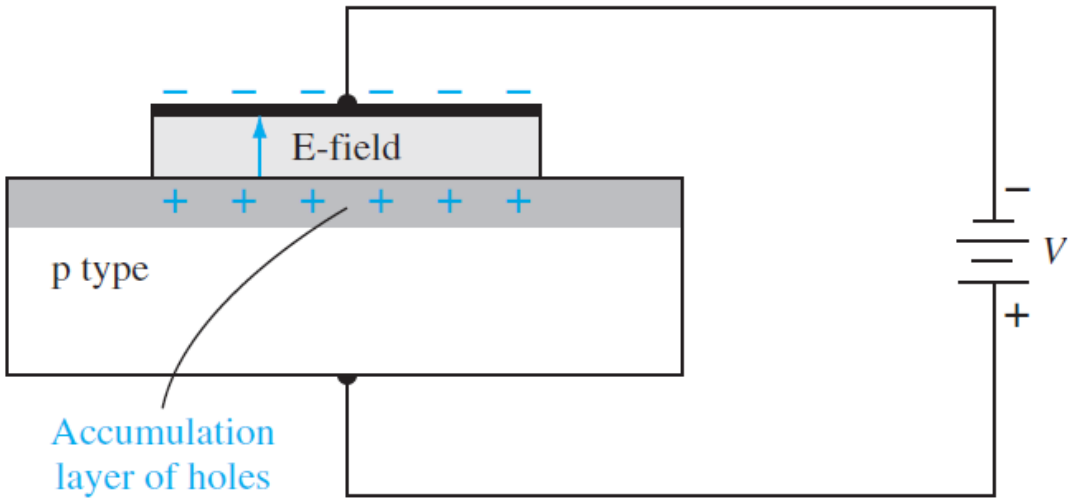
$$E = \frac{V}{d}$$



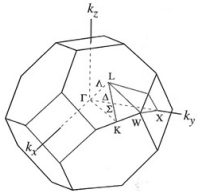




MOS with Negative Gate Bias



(c)



MOS with Positive Gate Bias

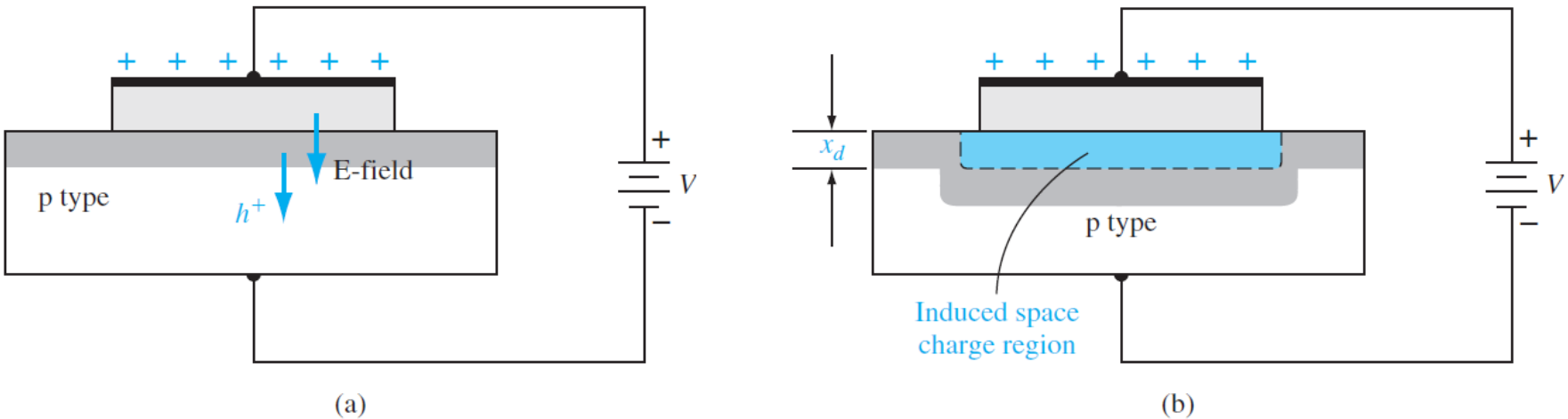
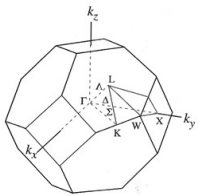
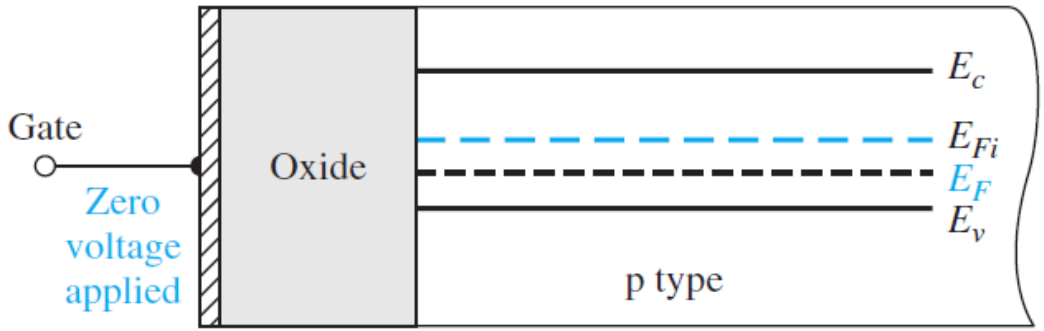


Figure 10.3 | The MOS capacitor with a moderate positive gate bias, showing (a) the electric field and charge flow and (b) the induced space charge region.

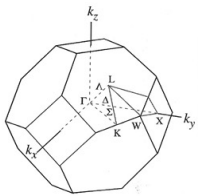


Band Diagram of a Ideal MOS capacitor

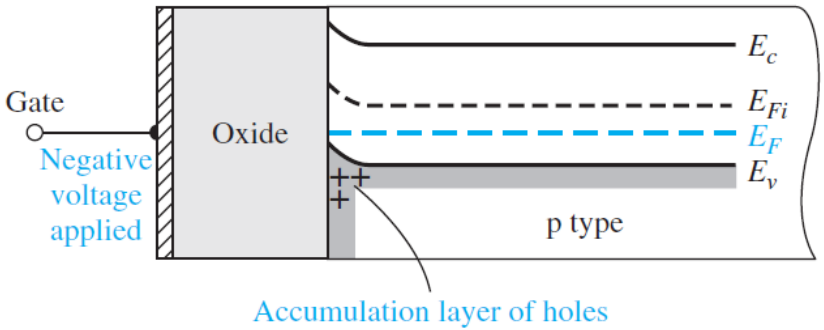


(a)

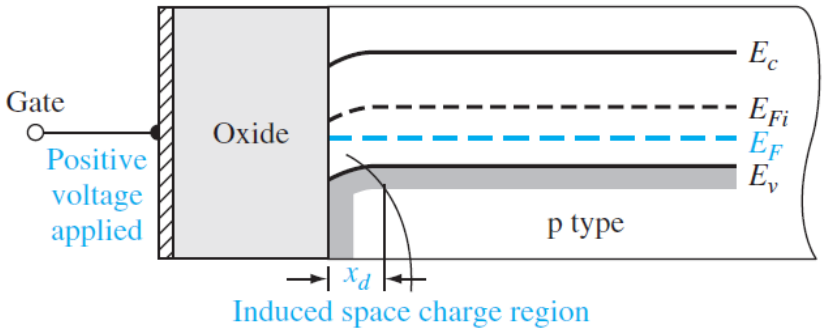
Potential, field, charge v.s distance?



Band Diagram of a Ideal MOS capacitor



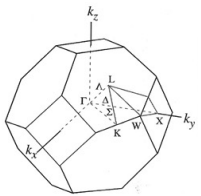
A Negative Gate Bias



A Positive Gate Bias

Potential, field, charge v.s distance?

$$\phi_s = \frac{1}{q}\left(E_{i,bulk} - E_{i,surf}\right)$$



Inversion Layer

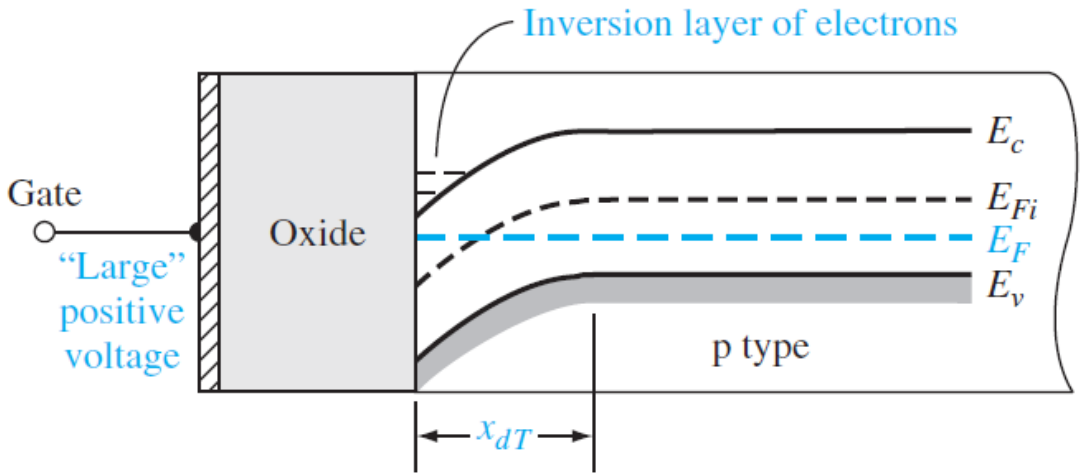
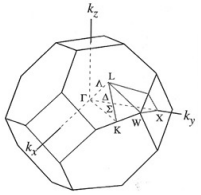


Figure 10.5 | The energy-band diagram of the MOS capacitor with a p-type substrate for a "large" positive gate bias.



MOS Capacitor on n-type Substrate

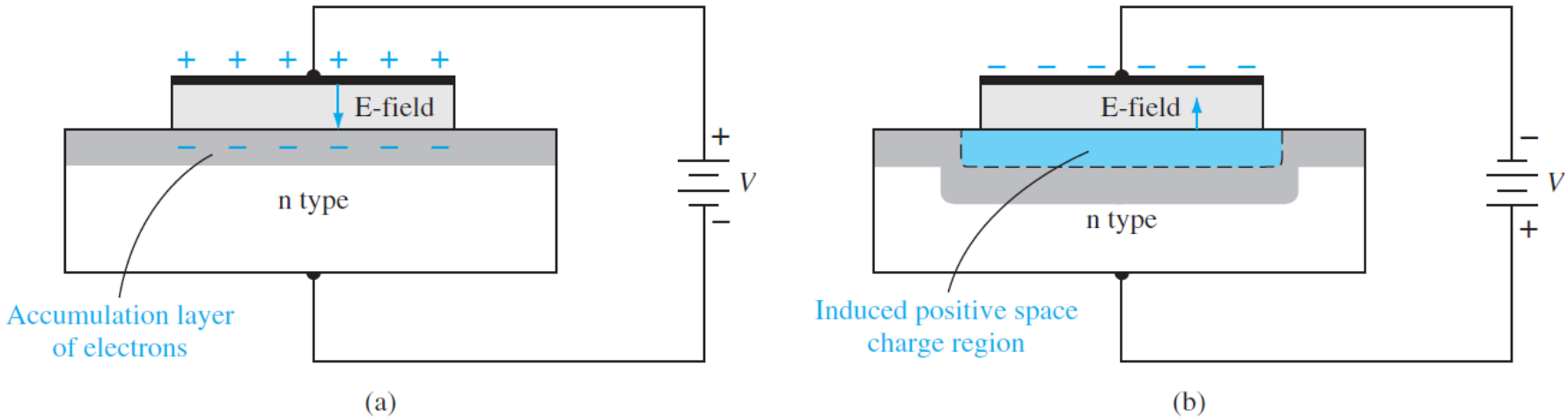
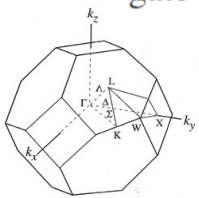
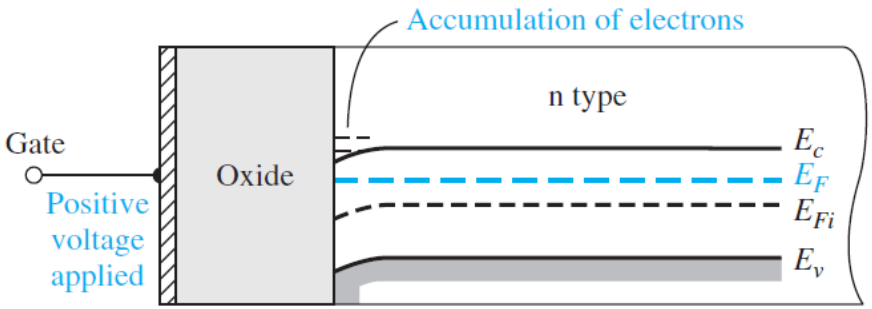
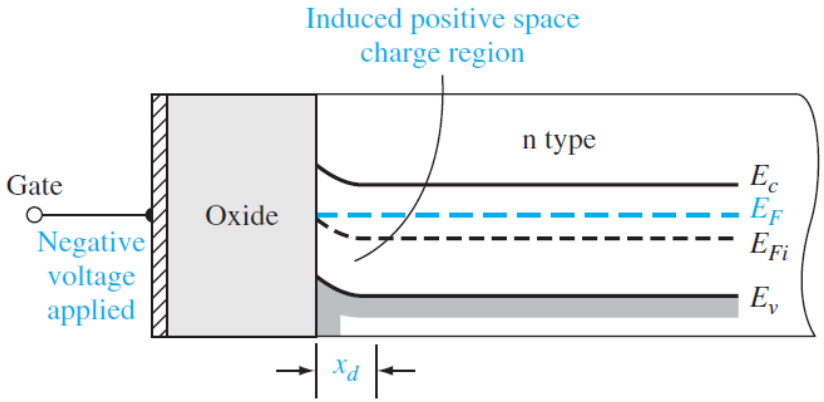


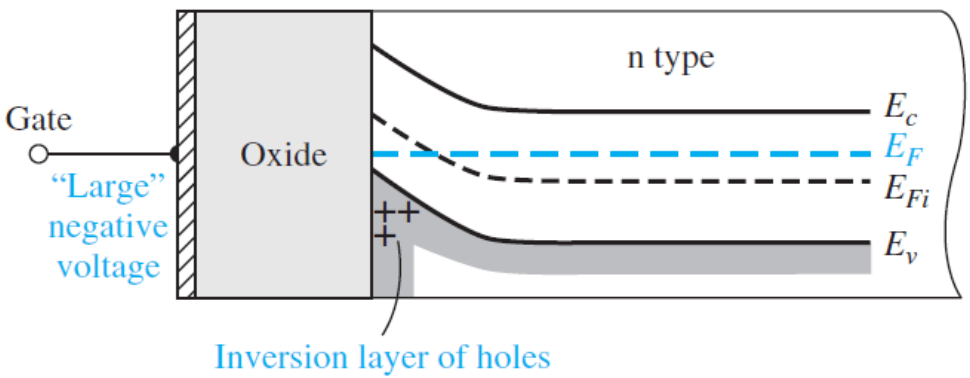
Figure 10.6 | The MOS capacitor with an n-type substrate for (a) a positive gate bias and (b) a moderate negative gate bias.

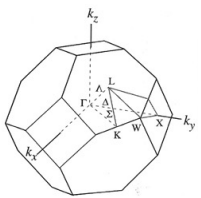


MOS Capacitor on n-type Substrate



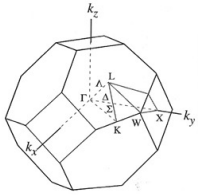




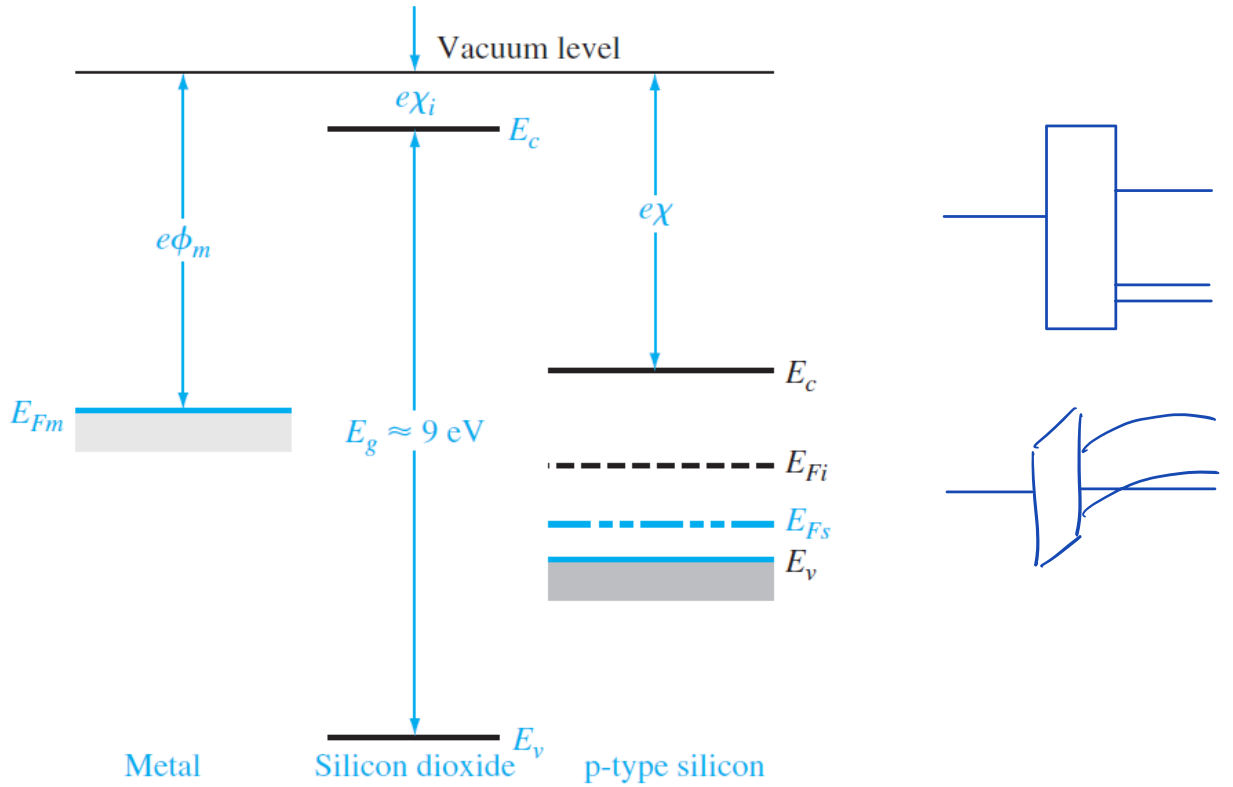


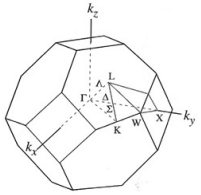
Two Important Reference Voltages

- Flat-band voltage, V_{FB} = voltage needed on gate to get E-field = 0 everywhere (flat bands). Note, this can be zero ("ideal" MOS), but generally depends on gate Φ_m or doping, qV_{FB} = ?
- Threshold voltage, V_T = voltage needed on gate to get electron concentration at Si/SiO_2 surface same as that of (majority) holes in the bulk. i.e. $\Phi_s(\text{inv}) = 2\Phi_F$ and Si surface is "inverted".



Work Function Difference

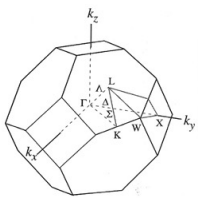


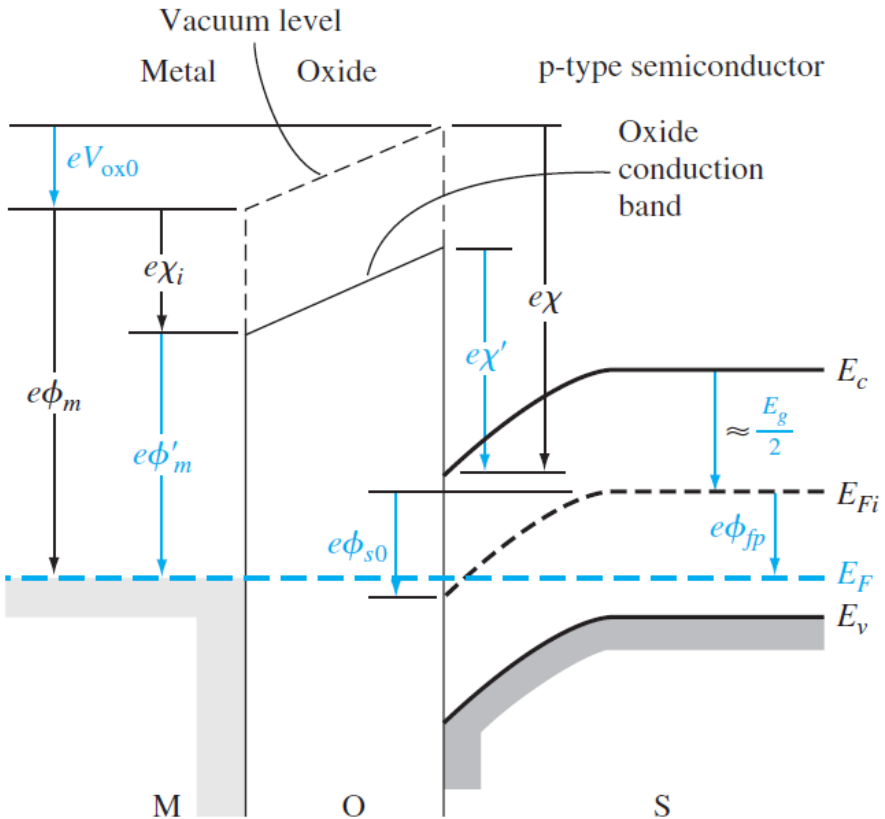


Work Function Difference

$$V_{ox0} + \phi_{s0} = -\left[\phi'_m - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)\right]$$

$$\phi_{ms} \equiv \left[\phi'_m - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)\right]$$

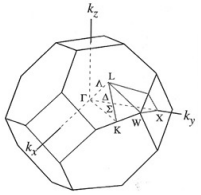




Example

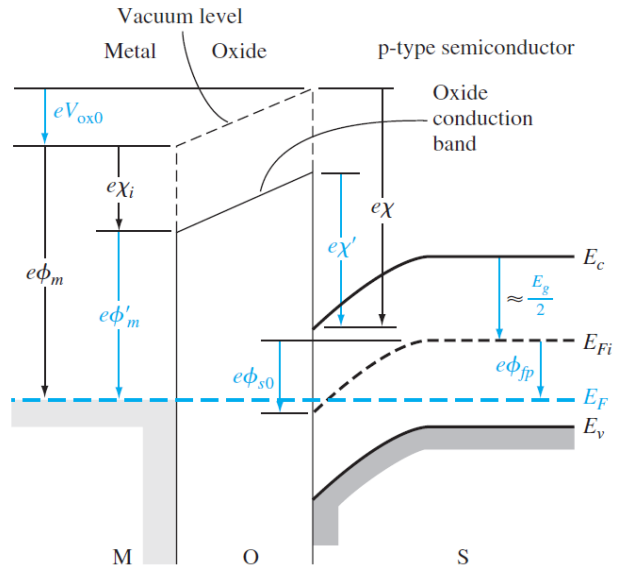
Objective: Determine the metal–semiconductor work function difference, ϕ_{ms}, for a given MOS system and semiconductor doping.

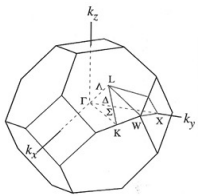
For an aluminum–silicon dioxide junction, $\phi'_m = 3.20$ V and, for a silicon–silicon dioxide junction, $\chi' = 3.25$ V. We may assume that $E_g = 1.12$ V. Let the p-type doping be $N_a = 10^{15}$ cm^{-3}.



Flat-band voltage

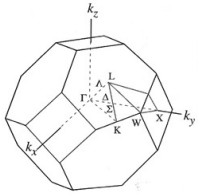
- Flat-band voltage, V_{FB} = voltage needed on gate to get E-field = 0 everywhere (flat bands). Note, this can be zero ("ideal" MOS), but generally depends on gate Φ_m or doping, V_{FB} = ?





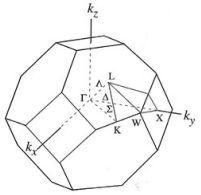
Voltage Applied on Gate

- In general, voltage applied on gate will be:
  - $V = V_{FB} + V_i + \emptyset_s$
  - $V_i = E_i d$: Voltage dropped across insulator
  - $\emptyset_s$ voltage dropped in the Silicon
- Q: What is the V_i when $V = V_{FB}$?



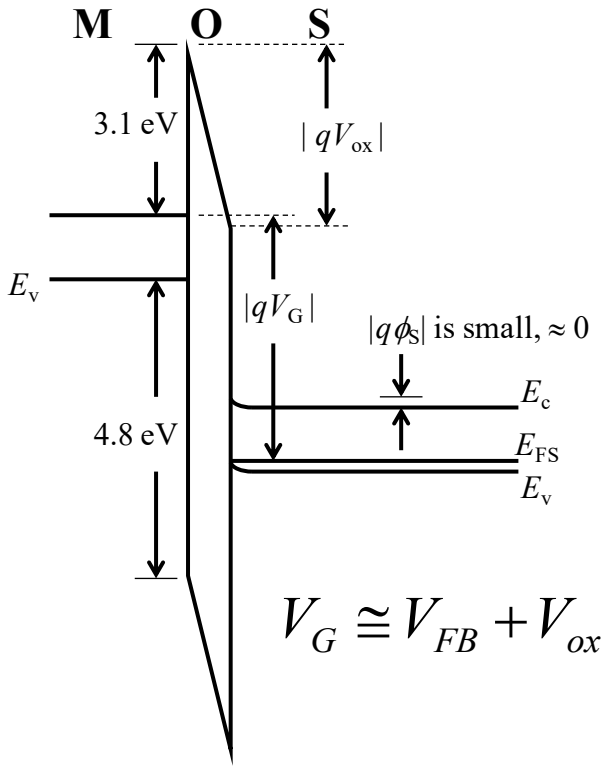
Three Regions of MOS Operation

- Accumulation ($V < V_{FB}$ for nMOS)
- Depletion ($V_{FB} < V < V_T$)
- Inversion ($V_T < V$)

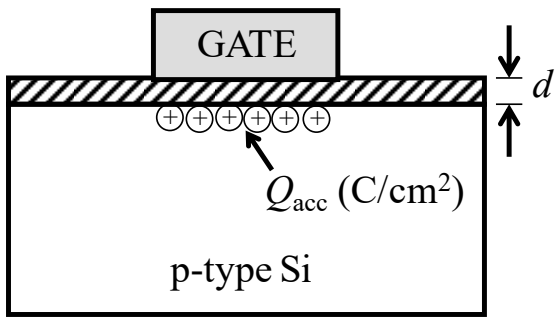


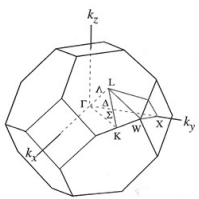
Accumulation

- $V < V_{FB}$, holes accumulate at the surface



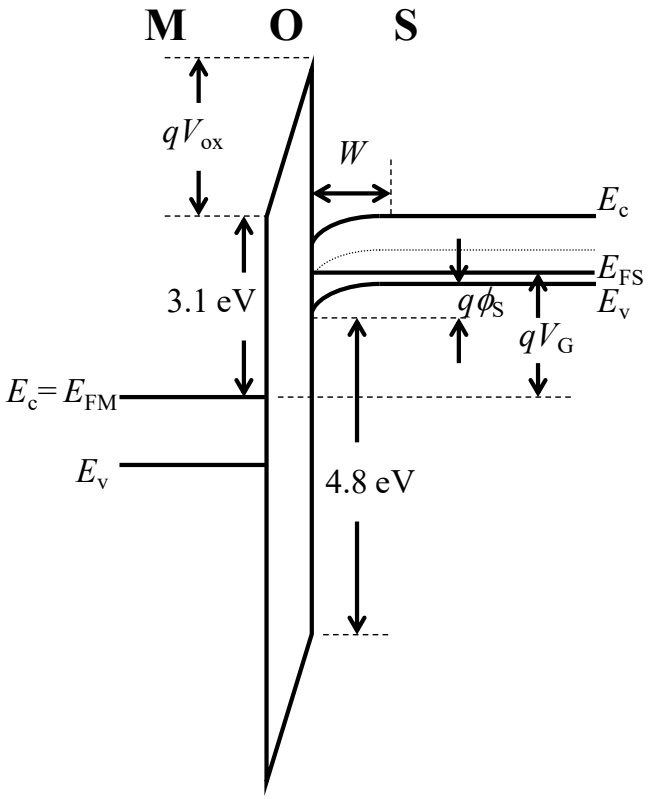
$$V_G \cong V_{FB} + V_{ox}$$

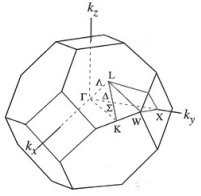




Depletion

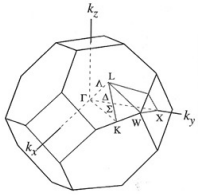
- $V_{FB} < V < V_T$, holes pushed back in substrate
  - Surface is depleted of mobile carriers
  - All surface charge is due to fixed dopant atoms
- Again, we apply depletion approximation we used for p-n diode: assume *abrupt* displaced charge (rectangular).





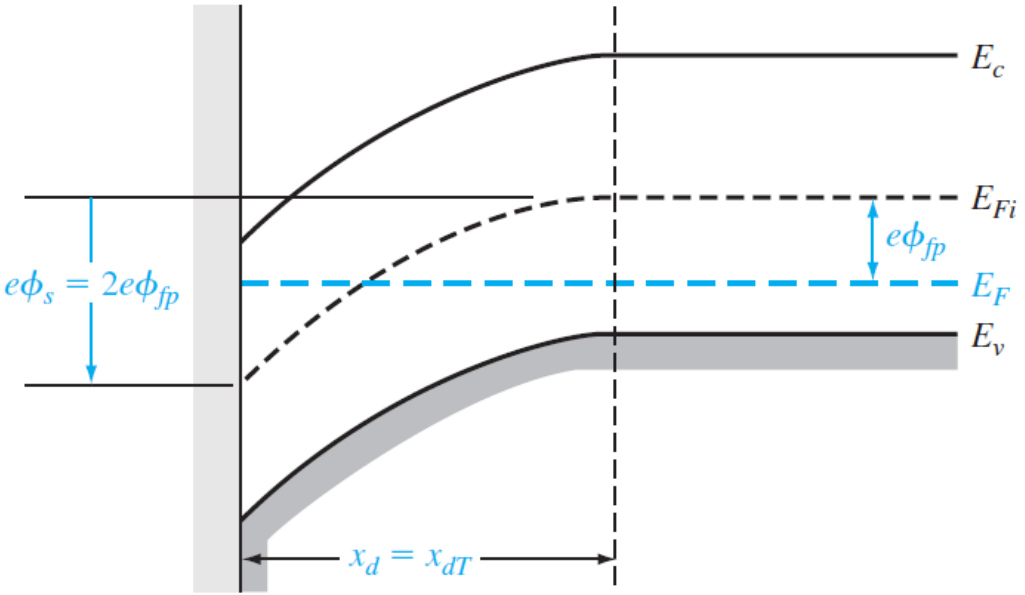
Depletion

- Charge density in depleted region: $\rho \cong -qN_A \quad (0 \le x \le W)$
- Poisson's equation in depleted region: $\frac{dE}{dx} = \frac{\rho}{\varepsilon_{Si}} \cong -\frac{qN_A}{\varepsilon_{Si}} \quad (0 \le x \le W)$
- Integrate twice (from bulk x = W to surface x = 0 to obtain surface Φ_s or depletion depth W:$\emptyset_s = \frac{qN_AW^2}{2\varepsilon_{Si}}$
- To find Φ_s as a function of gate V we need all voltage drops.
  - Across insulator: $V_i = E_i d$
  - Finally, $V_G = V_{FB} + \Phi_s + V_i$
- We can now solve from the surface potential vs. gate V:



Threshold Inversion Point

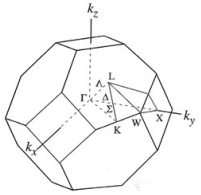
$$\phi_{fp} = V_t \ln\left(\frac{N_a}{n_i}\right)$$



$$x_d = \left(\frac{2\epsilon_s \phi_s}{eN_a}\right)^{1/2}$$

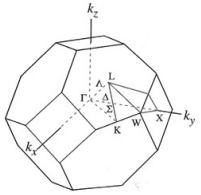
$$\phi_s = \frac{1}{q}\left(E_{i,bulk} - E_{i,surf}\right)$$

Figure 10.9 | The energy-band diagram in the p-type semiconductor at the threshold inversion point.



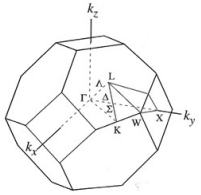
Inversion

- at $V > V_T$ (for NMOS), many electrons drawn to surface, which is now "inverted" vs. the p-doped substrate.
- Note:
  - Mobile charge anywhere is given by difference between E_C or E_V and Fermi level as, e.g. $n = N_C \exp(E_C - E_F)$
  - If E_F is close to E_C then lots of _____
  - If E_F is close to E_V then lots of _____
- At high gate V ($> V_T$) which band is closer to E_F?
- The condition for this onset is $\phi_S = 2\phi_F$



Inversion

- At inversion (on p-type sub.) the mobile surface charges are _____, such that numerically $n_{surf.} = N_{A,sub.}$
- Now we can calculate the threshold voltage, V_T:



Inversion

- For NMOS (electron inversion on p-type substrate):

$$V_G = V_{FB} + \varphi_S + V_i = V_{FB} + \varphi_S + \frac{\sqrt{2qN_A\varepsilon_{si}\varphi_S}}{C_i}$$

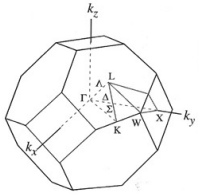
$$V_T = V_{FB} + 2\varphi_F + \frac{\sqrt{2qN_A\varepsilon_{Si}(2\varphi_F)}}{C_i}$$

$$\phi_F = \frac{kT}{q}\ln(N_A / n_i) > 0$$

- For PMOS (hole inversion on n-type substrate):

$$V_T = V_{FB} + 2\varphi_F - \frac{\sqrt{2qN_D\varepsilon_{Si}|2\varphi_F|}}{C_i}$$

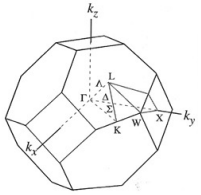
$$\varphi_F = -\frac{kT}{q}\ln(N_D / n_i) < 0$$



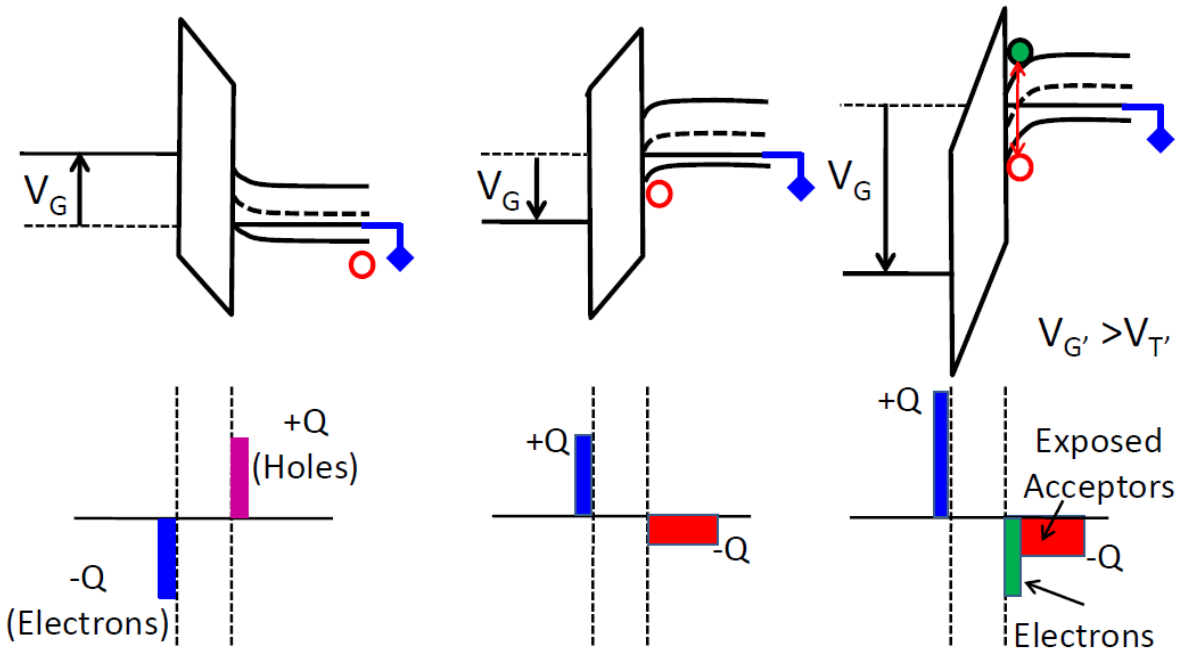
Example

Objective: Calculate the maximum space charge width for a given semiconductor doping concentration.

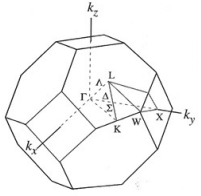
Consider silicon at $T = 300$ K doped to $N_a = 10^{16}$ cm^{-3}. The intrinsic carrier concentration is $n_i = 1.5 \times 10^{10}$ cm^{-3}.



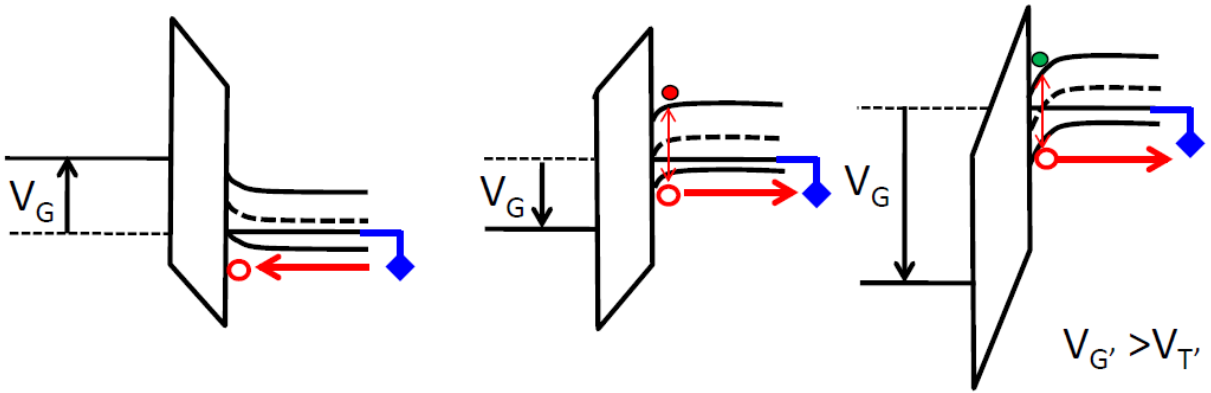
Where Do Charges Come From?



- Integrate charge to find potential.



Response Time

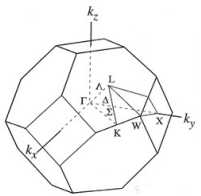


Dielectric Relaxation

$$\tau = \frac{\sigma}{\kappa_s \varepsilon_0}$$

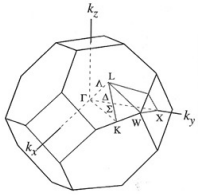
SRH Recombination-Generation

$$R = \frac{np - n_i^2}{\tau_n (p + p_1) + \tau_p (n + n_1)} \rightarrow \frac{-n_i}{\tau_n + \tau_p}$$





But how can we operate gate at today's clock frequency (~ 2 GHz!) if we can't generate minority carriers fast enough (> 100 Hz) ?



MOScap vs MOSFET

